

Inflation Dynamics - Computational Notebook

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Abstract

This book presents computation details of estimating inflation dynamics.

This notebook presents computation details of estimating inflation dynamics.

So far we have covered estimation of

1. GARCH
2. EGARCH
3. GJR-GARCH
4. SPARCH
5. Regime-switching GARCH
6. BEGE GARCH

State variable definitions:

1. π_t : inflation.
2. $\mathcal{F}_{t-1}$: filtration at time t.
3. SPF_t : professional forecast observed in $\mathcal{F}_{t-1}$.
4. $\hat{\pi}_t$: expected inflation $E[\pi_t \mid \mathcal{F}_{t-1}]$.
5. u_t : inflation innovation (residual), defined as $u_t := \pi_t - \hat{\pi}_t$.
6. u_t^+ : positive residual, defined as $u_t^+ := u_t \cdot \mathbf{1}(u_t > 0)$
7. u_t^- : negative residual, defined as $u_t^- := u_t \cdot \mathbf{1}(u_t \leq 0)$

1. INFLATION SUMMARY STATISTICS

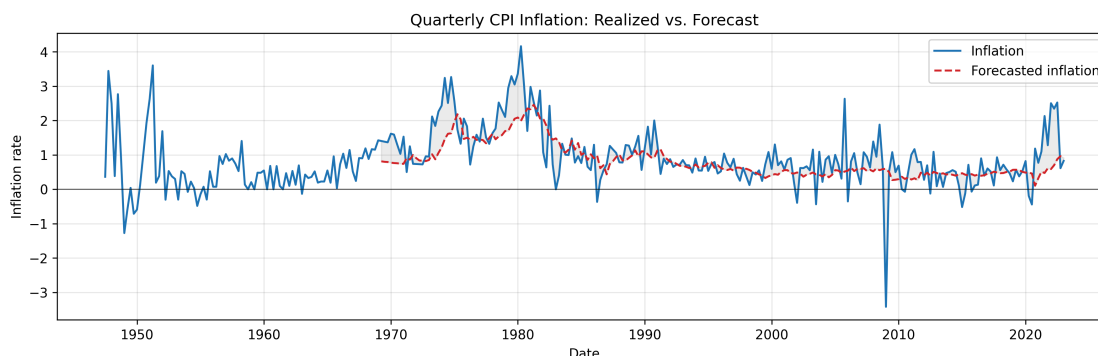


Figure 1: Quarterly Realized inflation and profesional forecast.

Notice that inflation was recorded from 1947 but profesional forecast started from 1969.

	Inflation	Forecasted inflation	Inflation shock
Date Start	1947Q2	1969Q1	1969Q1
Date End	2022Q4	2022Q4	2022Q4
#Observations	303	216	216

Table 1: Inflation statistics

1.a. The Number of Observations

To make log-likelihood values comparable across specifications, I use a common effective sample for all mean models and all volatility models.

1.a.i. Mean Process: Handling Unobserved Lags in AR/ARX:

In ARX models, lagged regressors (for example, SPF_{t-1}) are not observed at the very beginning of the raw sample.

Therefore, I start the estimation sample at the first date where both:

1. current forecast inflation (SPF_t), and
2. one-lag forecast inflation (SPF_{t-1})

are available.

For the quarterly data used here:

- forecast inflation (SPF_t) is available from **1969Q1** to **2022Q4** (216 observations),
- requiring one lag (SPF_{t-1}) shifts the usable start to **1969Q2**,
- so the trimmed estimation window is **1969Q2–2022Q4**, with **215 observations**.

This same trimmed sample is then used for all mean specifications (constant, ARX(1,1), ARX(2,1), etc.), so the residual series has the same length across models.

1.a.ii. Volatility Process: Handling Unobserved Initial σ_t^2 :

For GARCH-type recursion, the first few conditional variances are unobserved (for example, $\sigma_0^2, \sigma_{-1}^2$). I initialize these pre-sample variances using the model-implied unconditional variance:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \sum_i \alpha_i - \sum_j \beta_j}, \quad (1)$$

and set

$$\sigma_0^2 = \sigma_{-1}^2 = \dots = \bar{\sigma}^2. \quad (2)$$

Then I run the volatility recursion forward using the observed residuals from the common mean-model sample.

	Inflation	SPF	SPF_shock
Date Start	1969Q2	1969Q2	1969Q2
Date End	2022Q4	2022Q4	2022Q4
#Observations	215	215	215
Mean	0.9918	0.8320	0.1598
Median	0.7937	0.6296	0.1101
Std	0.8580	0.4969	0.6642
Min	-3.4170	0.1048	-4.0117
Max	4.1612	2.4566	2.1729
P5	-0.0681	0.3482	-0.7745
P25	0.5328	0.4748	-0.1669
P75	1.2913	1.0150	0.4617
P95	2.5883	2.0030	1.2581
Skewness	0.3550	1.3633	-0.7131
Kurtosis	3.8995	1.2053	7.3006
AC(1)	0.5981	0.9717	0.2764
AC(2)	0.5353	0.9473	0.1322
AC(4)	0.4708	0.8929	0.0881
AC(12)	0.2918	0.6949	0.0011

Table 2: Inflation statistics

2. MEAN PROCESS SPECIFICATION

Four types of mean processes are of interest:

- **Constant:** $\pi_{t+1} = SPF_t + \mu_{t+1}$
- **ARX(1,1):** $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,1):** $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,2)¹:** $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + \mu_{t+1}$

where μ_{t+1} is the residual term.

3. OLS RESULTS

I report OLS estimation results of three mean models with HAC standard error²:

ARX(1,1):

$$\hat{\pi}_{t+1} = 0.0720 + 0.2881 \pi_t + 0.7508 SPF_t \quad (3)$$

(0.087) (0.114) (0.168)

ARX(2,1):

$$\hat{\pi}_{t+1} = 0.0792 + 0.2793 \pi_t + 0.0728 \pi_{t-1} + 0.6661 SPF_t \quad (4)$$

(0.087) (0.102) (0.127) (0.257)

ARX(2,2):

$$\hat{\pi}_{t+1} = 0.0761 + 0.2880 \pi_t + 0.0799 \pi_{t-1} + 0.4720 SPF_t + 0.1793 SPF_{t-1} \quad (5)$$

(0.088) (0.109) (0.127) (0.440) (0.295)

¹In the GARCH family estimation exercise, we found that ARX(2,2) is always dominated by other three specifications. Thus, we stop estimating ARX(2,2) in BEGE estimation to saving computational resources.

²Standard errors are heteroskedasticity and autocorrelation robust (HAC) using 12 lags and without small sample correction.

4. GARCH

4.a. Model Specification

GARCH(p,q) model specification:

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i u_{t-i}^2 + \sum_{k=1}^q \beta_k \sigma_{t-k}^2 \quad (6)$$

5. GARCH ESTIMATION IN PYTHON'S arch PACKAGE: SAMPLE USE AND BACKCASTING

5.a. How arch Uses the Sample

For a GARCH(p, q) model with N observations and residuals ε_t :

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2 \quad (7)$$

arch uses **all N observations** in the log-likelihood — it does not drop the first $\max(p, q)$. The unobserved pre-sample values are filled in via **backcasting**: a single value $\hat{\sigma}_{bc}^2$ initializes all required pre-sample quantities,

$$\hat{\varepsilon}_0^2 = \hat{\varepsilon}_{-1}^2 = \dots = \hat{\sigma}_0^2 = \hat{\sigma}_{-1}^2 = \dots = \hat{\sigma}_{bc}^2 \quad (8)$$

5.b. The Four Standard Backcasting Methods

1. Sample (unconditional) variance. Equal weighting of squared residuals:

$$\hat{\sigma}_{bc}^2 = \frac{1}{N} \sum_{t=1}^N \varepsilon_t^2 \quad (9)$$

Default in EViews and MATLAB.

2. Model-implied long-run variance (variance targeting). Unconditional variance from current parameters:

$$\hat{\sigma}_{bc}^2 = \frac{\omega}{1 - \sum_i \alpha_i - \sum_j \beta_j} \quad (10)$$

Undefined when $\sum \alpha_i + \sum \beta_j \geq 1$.

5.c. Estimation Summary

5.d. Best Estimation Illustrations

6. GJR-GARCH SPECIFICATION

GJR GARCH model specification:

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i |u_{t-i}|^2 + \sum_{j=1}^o \gamma_j |u_{t-j}|^2 I[u_{t-j} < 0] + \sum_{k=1}^q \beta_k \sigma_{t-k}^2 \quad (11)$$

7. INFLATION DYNAMICS

To start the random search of initial mean parameters, I draw uniform samples from $(\mu - 2\sigma, \mu + 2\sigma)$ where μ and σ are mean and standard deviation from OLS regression. I also set the AR coefficient bound to avoid the AR process to explode. We have four types of mean processes.

Table 1: Parameter Bounds for Mean Process Specifications

Model	c	ρ_1	ρ_2	ϕ_1	ϕ_2
Constant	—	—	—	—	—
ARX(1,1)	$(\min \pi_t, \max \pi_t)$	$(-0.999, 0.999)$	—	$(-10, 10)$	—
ARX(2,1)	$(\min \pi_t, \max \pi_t)$	$(-1.999, 1.999)$	$(-0.999, 0.999)$	$(-10, 10)$	—
ARX(2,2)	$(\min \pi_t, \max \pi_t)$	$(-1.999, 1.999)$	$(-0.999, 0.999)$	$(-10, 10)$	$(-10, 10)$

7.a. II. GARCH Model Summarization

Table 2: Model selection by AIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	(1,1)	(2,1)	359.4710	(1,1)	(1,1)	368.2429
t	(1,1)	(2,1)	343.1431	(2,1)	(1,1)	344.6254
Skew t	(1,1)	(2,1)	345.1312	(2,1)	(1,1)	346.5791
GED	(1,1)	(2,1)	345.1925	(1,1)	(1,1)	347.6934
Mix of Normal	(1,1)	(2,1)	345.8222	(2,1)	(2,1)	343.5023

Table 3: Model selection by BIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	FC	(2,1)	381.4280	FC	(1,1)	383.4754
t	FC	(1,1)	367.8350	FC	(1,1)	367.1732
Skew t	FC	(1,1)	370.4242	FC	(1,1)	369.6649
GED	FC	(1,1)	371.8489	FC	(1,1)	370.9644
Mix of Normal	FC	(1,1)	373.7106	FC	(1,1)	371.0689

EGARCH model specification:

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \left(|e_{t-i}| - \sqrt{2/\pi} \right) + \sum_{j=1}^o \gamma_j e_{t-j} + \sum_{k=1}^q \beta_k \ln \sigma_{t-k}^2 \quad (12)$$

where $e_t = \varepsilon_t / \sigma_t$

8. BEGE DENSITY

BEGE density function $f_{BEGE}(\mu | p, n, \sigma_p, \sigma_n)$ is the function that calculates the density of the observation μ given the parameters $\{p, n, \sigma_p, \sigma_n\}$.

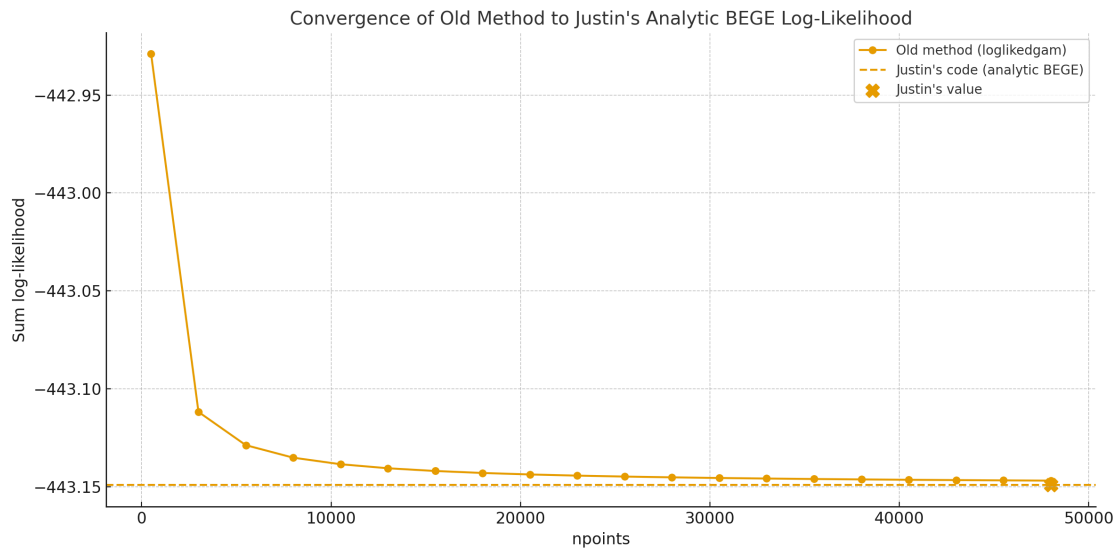
To compare Justin's analytic BEGE density code with the old numerical one, I conducted an experiment using synthetic data. I generated 200 independent observations from a standard normal distribution and fixed the BEGE parameters to

$$p = 5, \quad n = 7, \quad \sigma_p = 0.8, \quad \sigma_n = 1.2. \quad (13)$$

I computed the sum of log-likelihood using two approaches:

1. The *old BEGE function*, which approximates the density via numerical integration on a uniform grid, and whose accuracy depends strongly on the number of grid points; and
2. *Justin's analytic BEGE function*.

The old numerical method was evaluated over a wide range of grid resolutions (up to 50,000 points), and the resulting log-likelihood sums were plotted alongside the benchmark value computed from Justin's code.



Findings:

- The old numerical likelihood converges toward Justin's analytic likelihood as the number of grid points increases, but the convergence is slow and requires very fine grids.
- For small or moderate grid sizes, the numerical integration *overestimates* the log-likelihood. This explains why, for a given set of parameters, Justin's implementation produces a lower log-likelihood than the old code.
- Justin's analytic BEGE formula provides a more accurate and more computationally efficient evaluation.

9. BEGE GARCH FAMILY

9.a. Constant BEGE

The first model that I estimated is BEGE with invariant shape parameters, which is

$$\begin{aligned}\pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1} \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n\end{aligned}\tag{14}$$

where

$$\omega_p \sim \tilde{\Gamma}(\bar{p}, 1), \quad \omega_n \sim \tilde{\Gamma}(\bar{n}, 1).\tag{15}$$

$\bar{p}$ and $\bar{n}$ are unconditional shape parameters. The random search and bounds are set to be:

- $0.05 < \sigma_p, \sigma_n < 2$
- $0.1 < p, n < 10$

9.b. Symmetric BEGE

The model assumes different scaled parameters and different constants in GJR recursions for both components:

$$\begin{aligned}\pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n,\end{aligned}\tag{16}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1).\tag{17}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho, \phi^+, \phi^-)$:

$$\begin{aligned}p_t &= p_0 + \rho p_{t-1} + \frac{\phi^+}{2\sigma_p^2} (u_{t-1}^+)^2 + \frac{\phi^-}{2\sigma_p^2} (u_{t-1}^-)^2, \\ n_t &= n_0 + \rho n_{t-1} + \frac{\phi^+}{2\sigma_n^2} (u_{t-1}^+)^2 + \frac{\phi^-}{2\sigma_n^2} (u_{t-1}^-)^2\end{aligned}\tag{18}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho < 0.999$,
- $10^{-5} < \phi^+, \phi^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

9.c. Bad and Good Symmetric GARCH Model

Here n_t, p_t is generated recursively with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p, \phi_n)$:

$$\begin{aligned}p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p}{2\sigma_p^2} (u_{t-1})^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n}{2\sigma_n^2} (u_{t-1})^2\end{aligned}\tag{19}$$

I have set the constraints below.

- $\rho + \phi < 1$ for both good and bad shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$.

- $\max\{p_t, n_t\} < 200$

9.d. Inflation and Deflation Model

Here n_t, p_t is generated recursively with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_n^-)$:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} (u_{t-1}^+)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^-}{2\sigma_n^2} (u_{t-1}^-)^2 \end{aligned} \quad (20)$$

According to the emails, I have set the constraints below.

- $\rho_p + \frac{\phi_p^+}{2} < 1$ and $\rho_n + \frac{\phi_n^-}{2} < 1$.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$.
- $\max\{p_t, n_t\} < 200$

9.e. Full BEGE

BEGE GARCH has mean process and higher:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n, \end{aligned} \quad (21)$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \quad (22)$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^-)$:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} (u_{t-1}^+)^2 + \frac{\phi_p^-}{2\sigma_p^2} (u_{t-1}^-)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2} (u_{t-1}^+)^2 + \frac{\phi_n^-}{2\sigma_n^2} (u_{t-1}^-)^2 \end{aligned} \quad (23)$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho_p, \rho_n < 0.999$,
- $10^{-5} < \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

According to the emails, I have set the constraints below.

- $\rho + \frac{\phi^+}{2} + \frac{\phi^-}{2} < 1$ for both BE and GE shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$ where $\text{Var}(\pi_t) = 0.75$ is the unconditional variance of inflation.
- $\max\{p_t, n_t\} < 200$.

10. RESULT SUMMARY

Overall, most of the estimations with random initial values converge. I randomly sampled 100,000 starting values for the constant BEGE model, and sampled 10,000 for symmetric BEGE because it is more time consuming. I report the best estimation below.